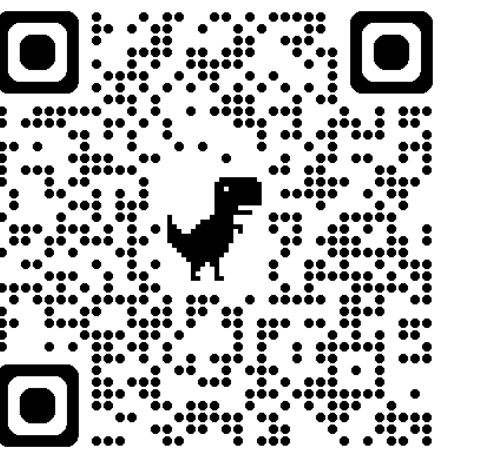
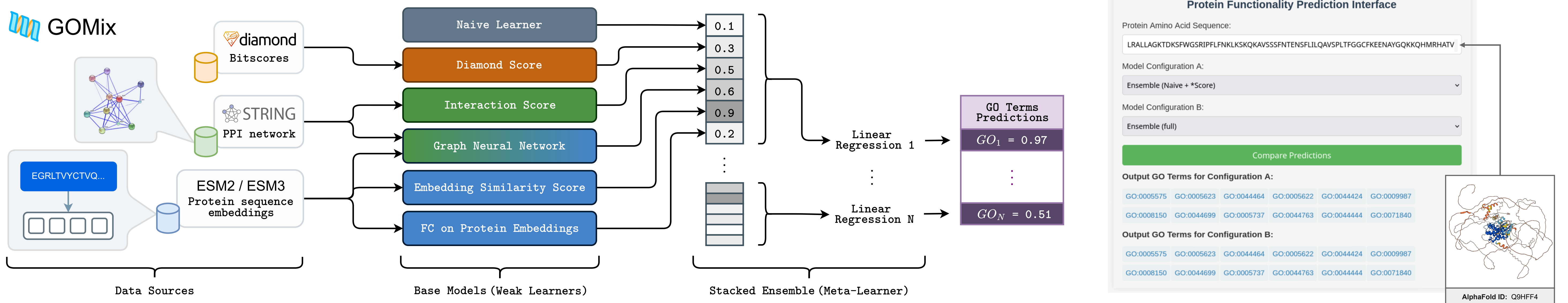


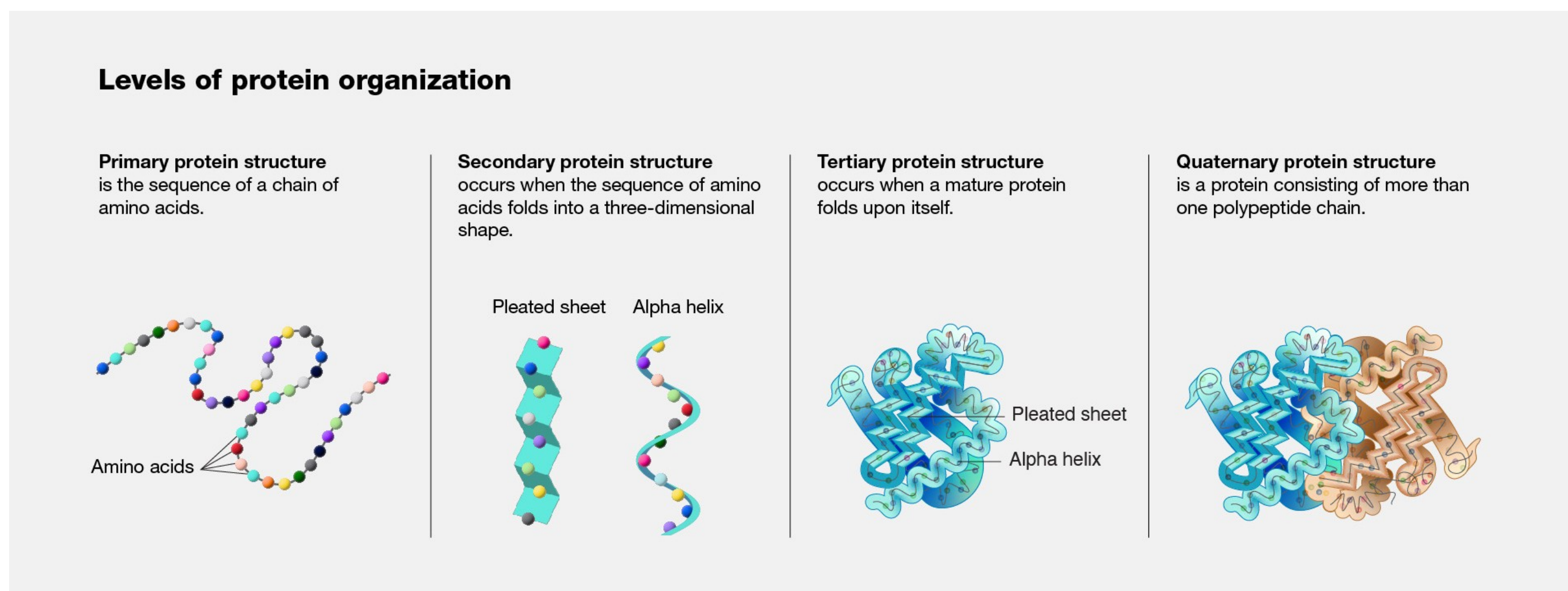


Method Overview



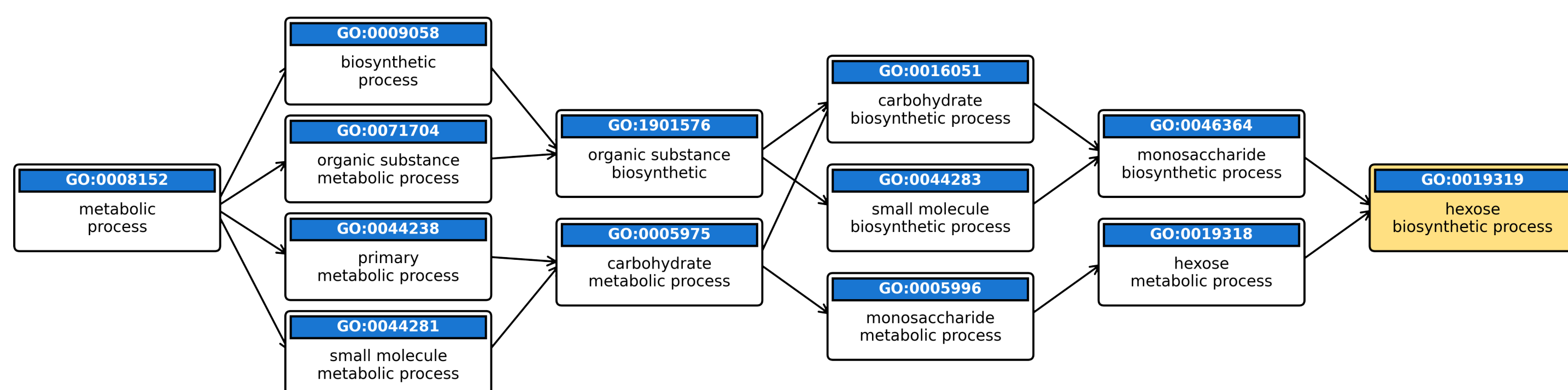
Preliminaries

- Every living organism is built from **cells**—tiny factories that carry out all the fundamental processes of life; e.g., an adult human has 30–40T cells.
- Proteins** are the molecules that do the actual work inside the cells, built according to the instructions encoded in their **genes**; e.g., each human cell contains roughly 1–10B proteins belonging to 10–12K different types.
- Each protein can have **one or more functions** in the cell; the **Gene Ontology (GO)** is a global catalog ($\sim 45K$ entries, 3 sub-ontologies).
 - Molecular Function Ontology (MFO)**—what the protein does (e.g., transport molecules, cut molecules, transmit signals, catalyze reactions).
 - Biological Process Ontology (BPO)**—what task it supports (e.g., transport oxygen through the bloodstream, build and repair muscles).
 - Cellular Component Ontology (CCO)**—where it works in the cell.
 - “Proteins carry out countless molecular functions that collectively drive physiological processes (e.g., breathing, digestion, movement) and contribute to phenotypic traits (e.g., eye or hair color).”
- Proteins are **sequences of up to hundreds of amino acids** (from 20 types).
 - The amino-acid sequence determines the 3D structure and dynamics of the protein, which in turn defines its function.



Task

- Motivations**
 - Due to ongoing genome sequencing projects, we are inundated with large amounts of genomic sequence data from thousands of species; **$\sim 250M$ protein sequences registered on the UniProtKB database!**
 - However, **we still do not understand what many individual proteins do.**
 - Conducting laboratory experiments would be extremely expensive and time-consuming; thousands of US dollars per protein in core facilities.
- Protein Function Prediction (PFP)**
 - Automatic inference of a protein’s role by mapping its amino-acid sequence to one or more valid GO terms.
 - It helps to understand how cells operate and enables practical applications, from identifying disease mechanisms to developing new therapies that target or correct malfunctioning proteins.



GOMix

- An **open-source ensemble framework** that combines multiple prediction strategies to achieve robust PFP for **unseen proteins**.
- In our demo paper, we mount **seven base learners**...
 - Naive**. The probability of a GO term is its frequency in the training data annotation (i.e., protein independent).
 - Diamond**. If many proteins with sequences similar to the input run a function, the input likely does too. **Diamond alignment tool**.
 - Interaction**. If proteins that interact with the input run a function, the input likely does too. **STRING protein-protein interaction network**.
 - Embedding Similarity**. If proteins with similar embeddings run a function, the input likely does too. Protein language models: ESM2 (15B) and ESM3-open (1.4B). Embeddings encapsulate 2D and 3D structure, sequence homology, and physicochemical properties.
 - Fully Connected Model**. Seven-layer MLP network ($\sim 220M$ parameters) on ESM3 embeddings with $|GO\ terms|$ output nodes.
 - Graph Neural Network**. Two-layer GraphSAGE model ($\sim 795M$ parameters) on **STRING**. ESM3 node initialization. Graph-level task.
- ... Whose predictions are utilized by a **meta-learner** to determine a score for each target GO term within the training data.
 - One linear regressor per GO term, trained to modulate the relative contribution of each base learner.

Experimental Setup

- CAFA-compliant dataset**. Pre-2016 annotations for training (65K proteins, 80/20 for base/meta learners), Jan–Oct 2016 for testing (2K).
- Metrics**. Because models output confidence scores rather than binary labels, we sweep thresholds over **[0, 1]** with a step of 0.01 and calculate: F_{max} (maximum F1 attainable), S_{min} (minimum semantic distance between actual and predicted GO labels), AUPR.

Results

- Competitive or superior performance w.r.t. previous close solutions.
- Reduced ensemble. 1–5 achieves generalization close to 1–7 (full), with substantial efficiency gains—only 389 seconds for test set inference.

Method	F_{max} ($\uparrow$)			S_{min} ($\downarrow$)			AUPR ($\uparrow$)		
	MFO	BPO	CCO	MFO	BPO	CCO	MFO	BPO	CCO
◇ Naive	0.304	0.318	0.605	12.112	38.890	9.646	0.146	0.214	0.542
◇ Diamond Score	0.550	0.440	0.625	8.617	34.027	7.945	0.375	0.257	0.395
◇ Interaction Score	0.406	0.421	0.644	11.214	35.911	8.175	0.332	0.352	0.626
◇ Embedding Similarity Score (ESM-2)	0.579	0.471	0.705	8.698	33.631	7.513	0.549	0.425	<u>0.747</u>
◇ Embedding Similarity Score (ESM-3)	0.599	<u>0.491</u>	0.722	8.056	30.413	6.933	0.598	0.442	0.766
◇ FC on protein embeddings	0.531	0.450	0.690	9.734	34.859	7.933	0.440	0.361	0.700
◇ GNN on PPI & embeddings	0.435	0.361	0.606	11.108	38.407	8.914	0.336	0.233	0.545
UDSMProt	0.582	0.475	0.697	8.787	33.615	7.618	0.548	0.422	0.728
DeepText2GO	0.627	0.441	0.694	<u>5.240</u>	<u>17.713</u>	4.531	0.605	0.336	0.729
GOLabeler	0.580	0.370	0.687	5.077	15.177	<u>5.518</u>	0.546	0.225	0.700
DeepGOPlus	0.585	0.474	0.699	8.824	33.576	7.693	0.536	0.407	0.726
PANDA2	0.598	0.478	0.709	9.670	40.229	9.558	0.564	<u>0.436</u>	0.744
GOMix: Naive + *Score	0.601	0.498	<u>0.728</u>	8.217	29.620	6.098	<u>0.604</u>	0.444	<u>0.747</u>
GOMix: full	<u>0.604</u>	0.498	0.732	8.251	29.850	6.048	0.601	<u>0.436</u>	0.738

$\uparrow$ The higher, the better $\downarrow$ The lower, the better $\diamond$ GOMIX components

Table 1: Test set performance comparison. Bold and underline denote the best and second-best scores.